ENV_SETUP_GUIDE.md — Every Key You Need + How to Collect Each One

Purpose: Complete `.env.local` template for the School/Madrasa SaaS platform + step-by-step guide on *where* each key comes from, *how* to generate it, *how much* it costs, and *when* you need it (MVP vs later phase).

Audience: You (Famies) — for your own setup. Also pass to Claude Code so it knows exactly which variable names to reference in code.

📋 Quick Overview — Which Services, Which Phase

Phase	Service	Purpose	Cost	Required?
Phase 0	Supabase	DB + Auth + Storage	Free → $25/mo	✅ Must
Phase 0	Vercel	Hosting	Free → $20/mo	✅ Must
Phase 0	Resend	Transactional email	Free (3k/mo)	✅ Must
Phase 1	SSL Wireless SMS	Bulk SMS (BD)	~৳0.25/SMS	✅ Must
Phase 2	SSLCommerz	Payment gateway	2.5% per txn	✅ Must
Phase 2	bKash PGW	Mobile payment	1.85% per txn	✅ Recommended
Phase 2	Nagad PGW	Mobile payment	1.75% per txn	⚠️ Optional
Phase 3	Firebase FCM	Push notifications	Free	✅ Must
Phase 3	Cloudinary	Image optimization	Free (25GB)	⚠️ Optional

Phase	Service	Purpose	Cost	Required?
Phase 4	WhatsApp Business API	WhatsApp messaging	~৳0.50/msg	⚠️ Optional
Phase 5	Anthropic API	AI features (Claude)	Pay per use	⚠️ Optional
Phase 5	OpenAI	Backup AI (optional)	Pay per use	❌ Optional
Phase 6	Sentry	Error tracking	Free (5k events)	✅ Recommended
Phase 6	PostHog	Product analytics	Free (1M events)	✅ Recommended
Phase 6	Upstash Redis	Caching + rate limit	Free (10k/day)	⚠️ Optional
Phase 7	Stripe	International payment	2.9% + $0.30	❌ Only if global

Total monthly cost for MVP (~500 students): ~$30-50/month **Total for scale (~10,000 students across 20 schools):** ~$200-400/month

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🗂️ Complete `.env.local` Template

Copy this entire block into a file named `.env.local` in your Next.js project root. Then fill in each value by following the steps below.

```
# ==========================================================================

# CORE — Next.js & App Config

# ==========================================================================

NEXT_PUBLIC_APP_URL=http://localhost:3000
```

```
NEXT_PUBLIC_APP_NAME="Shikkha Platform"

NEXT_PUBLIC_APP_SHORT_NAME="Shikkha"

NODE_ENV=development

# JWT secret for custom signed tokens (generate: openssl rand -base64 32)

JWT_SECRET=

# Session encryption key (generate: openssl rand -hex 32)

SESSION_SECRET=

# ==========================================================================

# SUPABASE — Database + Auth + Storage + Realtime

# ==========================================================================

NEXT_PUBLIC_SUPABASE_URL=

NEXT_PUBLIC_SUPABASE_ANON_KEY=

SUPABASE_SERVICE_ROLE_KEY=

SUPABASE_JWT_SECRET=

SUPABASE_DB_PASSWORD=

SUPABASE_PROJECT_REF=

# ==========================================================================

# VERCEL — Hosting (auto-filled in Vercel dashboard; add only for local mimicry)
```

```
#
==========================================================================
===

VERCEL_URL=

VERCEL_ENV=development

#
==========================================================================
===

# EMAIL — Resend (transactional email)

#
==========================================================================
===

RESEND_API_KEY=

RESEND_FROM_EMAIL=noreply@shikkha.app

RESEND_FROM_NAME="Shikkha Platform"

RESEND_REPLY_TO=support@shikkha.app

#
==========================================================================
===

# SMS — SSL Wireless (primary BD SMS provider)

#
==========================================================================
===

SSL_SMS_API_TOKEN=

SSL_SMS_SID=

SSL_SMS_URL=https://smsplus.sslwireless.com/api/v3/send-sms

SSL_SMS_SENDER_ID=
```

```
# SMS — Alpha SMS (fallback)

ALPHA_SMS_API_KEY=

ALPHA_SMS_SENDER_ID=

ALPHA_SMS_URL=https://api.sms.net.bd/sendsms

# SMS — BulkSMSBD (secondary fallback)

BULKSMS_BD_API_KEY=

BULKSMS_BD_SENDER_ID=

# ==========================================================================

# PAYMENT — SSLCommerz (primary payment gateway, covers all cards + MFS)

# ==========================================================================

SSLCOMMERZ_STORE_ID=

SSLCOMMERZ_STORE_PASSWORD=

SSLCOMMERZ_SANDBOX=true

SSLCOMMERZ_SUCCESS_URL=${NEXT_PUBLIC_APP_URL}/api/payment/sslcommerz/success

SSLCOMMERZ_FAIL_URL=${NEXT_PUBLIC_APP_URL}/api/payment/sslcommerz/fail

SSLCOMMERZ_CANCEL_URL=${NEXT_PUBLIC_APP_URL}/api/payment/sslcommerz/cancel

SSLCOMMERZ_IPN_URL=${NEXT_PUBLIC_APP_URL}/api/payment/sslcommerz/ipn

# PAYMENT — bKash (direct integration for lower fees)

BKASH_APP_KEY=
```

```
BKASH_APP_SECRET=

BKASH_USERNAME=

BKASH_PASSWORD=

BKASH_BASE_URL=https://tokenized.sandbox.bka.sh/v1.2.0-beta

BKASH_CALLBACK_URL=${NEXT_PUBLIC_APP_URL}/api/payment/bkash/callback

# PAYMENT — Nagad (optional, direct integration)

NAGAD_MERCHANT_ID=

NAGAD_MERCHANT_NUMBER=

NAGAD_PG_PUBLIC_KEY=

NAGAD_MERCHANT_PRIVATE_KEY=

NAGAD_BASE_URL=https://api.mynagad.com/api/dfs

NAGAD_CALLBACK_URL=${NEXT_PUBLIC_APP_URL}/api/payment/nagad/callback

# PAYMENT — Stripe (only if serving international schools)

STRIPE_SECRET_KEY=

STRIPE_WEBHOOK_SECRET=

NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY=

# ==========================================================================

# PUSH NOTIFICATIONS — Firebase Cloud Messaging (FCM)

# ==========================================================================
```

```
NEXT_PUBLIC_FIREBASE_API_KEY=

NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN=

NEXT_PUBLIC_FIREBASE_PROJECT_ID=

NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET=

NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID=

NEXT_PUBLIC_FIREBASE_APP_ID=

NEXT_PUBLIC_FIREBASE_VAPID_KEY=

FIREBASE_ADMIN_PROJECT_ID=

FIREBASE_ADMIN_CLIENT_EMAIL=

FIREBASE_ADMIN_PRIVATE_KEY=

# ==========================================================================

# IMAGE STORAGE — Cloudinary (optional; Supabase Storage is the default)

# ==========================================================================

CLOUDINARY_CLOUD_NAME=

CLOUDINARY_API_KEY=

CLOUDINARY_API_SECRET=

CLOUDINARY_UPLOAD_PRESET=

# ==========================================================================

# WHATSAPP — Meta Cloud API (for parent communication)
```

```
#
==========================================================================
===

WHATSAPP_PHONE_NUMBER_ID=

WHATSAPP_BUSINESS_ACCOUNT_ID=

WHATSAPP_ACCESS_TOKEN=

WHATSAPP_VERIFY_TOKEN=

WHATSAPP_APP_SECRET=

#
==========================================================================
===

# AI — Anthropic Claude (report comments, dropout risk, smart suggestions)

#
==========================================================================
===

ANTHROPIC_API_KEY=

ANTHROPIC_MODEL=claude-sonnet-4-6

# AI — OpenAI (backup/fallback)

OPENAI_API_KEY=

OPENAI_MODEL=gpt-4o-mini

#
==========================================================================
===

# ERROR TRACKING — Sentry
```

```
#
==========================================================================
===

SENTRY_AUTH_TOKEN=

NEXT_PUBLIC_SENTRY_DSN=

SENTRY_ORG=

SENTRY_PROJECT=shikkha-platform

#
==========================================================================
===

# ANALYTICS — PostHog

#
==========================================================================
===

NEXT_PUBLIC_POSTHOG_KEY=

NEXT_PUBLIC_POSTHOG_HOST=https://app.posthog.com

#
==========================================================================
===

# CACHING + RATE LIMITING — Upstash Redis

#
==========================================================================
===

UPSTASH_REDIS_REST_URL=

UPSTASH_REDIS_REST_TOKEN=

#
==========================================================================
===
```

MAPS — Google Maps (for school location picker, transport routes)

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NEXT_PUBLIC_GOOGLE_MAPS_API_KEY=

==========

CRON / SCHEDULED JOBS — Vercel Cron or Upstash QStash

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CRON_SECRET=

QSTASH_TOKEN=

QSTASH_CURRENT_SIGNING_KEY=

QSTASH_NEXT_SIGNING_KEY=

==========

FILE UPLOAD — max sizes (no keys needed, just config)

==========

MAX_UPLOAD_SIZE_MB=10

MAX_PROFILE_IMAGE_SIZE_MB=2

```
# ==========================================================================

# FEATURE FLAGS

# ==========================================================================

NEXT_PUBLIC_ENABLE_AI_FEATURES=false

NEXT_PUBLIC_ENABLE_MADRASA_MODULE=true

NEXT_PUBLIC_ENABLE_WHATSAPP=false

NEXT_PUBLIC_ENABLE_BIOMETRIC=false

NEXT_PUBLIC_ENABLE_LMS=false

# ==========================================================================

# ADMIN (super admin bootstrap — only for first deploy)

# ==========================================================================

SUPER_ADMIN_EMAIL=

SUPER_ADMIN_INITIAL_PASSWORD=
```

🔑 Step-by-Step: How to Collect Every Single Key

1️⃣ SUPABASE (Most Important — Do This First)

Kaj: PostgreSQL database, Auth, File storage, Realtime subscriptions, Row Level Security.

Cost: Free tier includes 500MB DB + 1GB storage + 50k monthly active users. $25/mo Pro = 8GB DB + 100GB storage.

Steps:

1. Go to https://supabase.com
2. Click **"Start your project"** → Sign up with GitHub (recommended) or email
3. Click **"New Project"**
4. Fill:
   - **Name:** `shikkha-platform-dev` (for dev) / `shikkha-platform-prod` (for production)
   - **Database Password:** Generate a strong one — *save this somewhere safe immediately*
   - **Region:** `Singapore (ap-southeast-1)` — closest to Bangladesh
   - **Pricing plan:** Free (upgrade to Pro when you have paying schools)
5. Wait ~2 minutes for provisioning
6. Once ready, go to **Settings** → **API**:
   - Copy **Project URL** → `NEXT_PUBLIC_SUPABASE_URL`
   - Copy **anon / public key** → `NEXT_PUBLIC_SUPABASE_ANON_KEY`
   - Copy **service_role key** (⚠️ secret, never expose to frontend) → `SUPABASE_SERVICE_ROLE_KEY`
7. Go to **Settings** → **Database**:
   - Copy **Database password** → `SUPABASE_DB_PASSWORD`
   - Note the **Project Reference ID** (in the URL, e.g., `xyzabc123`) → `SUPABASE_PROJECT_REF`
8. Go to **Settings** → **API** → **JWT Settings**:
   - Copy **JWT Secret** → `SUPABASE_JWT_SECRET`

Tip: Create two separate Supabase projects — one for dev, one for production. Never mix.

2️⃣ VERCEL (Hosting)

Kaj: Host the Next.js app with edge functions + auto-SSL + CI/CD on git push.

Cost: Free (Hobby) for personal projects, $20/mo (Pro) for team/commercial use.

Steps:

1. Go to https://vercel.com
2. Sign up with GitHub
3. Click **"Add New → Project"**
4. Import your GitHub repo (we'll create this in Phase 0)

5. Before deploying, add all `.env.local` variables in **Environment Variables** section
6. Click **Deploy**
7. `VERCEL_URL` and `VERCEL_ENV` are auto-injected — no manual collection needed

Custom domain: Settings → Domains → Add `shikkha.app` (or whatever you buy). Vercel handles SSL automatically.

3️⃣ RESEND (Transactional Email)

Kaj: Send password reset, welcome emails, invoice PDFs, monthly reports via email.

Cost: Free = 3,000 emails/month + 100/day. $20/mo = 50k emails.

Steps:

1. Go to **https://resend.com**
2. Sign up → Verify email
3. Go to **API Keys** → **Create API Key**
4. Name: `shikkha-platform-prod`, Permission: `Sending access`
5. Copy the key (starts with `re_`) → `RESEND_API_KEY`
6. Go to **Domains** → **Add Domain**
7. Enter your domain (e.g., `shikkha.app`)
8. Resend gives you DNS records (SPF, DKIM, DMARC) — add these to your domain registrar (Namecheap/GoDaddy/Cloudflare)
9. Wait ~15 minutes → Click **Verify**
10. Once verified, set:
    - `RESEND_FROM_EMAIL=noreply@shikkha.app`
    - `RESEND_FROM_NAME="Shikkha Platform"`

Tip: Until your domain is verified, use `onboarding@resend.dev` for testing.

4️⃣ SSL WIRELESS SMS (Primary BD SMS Provider)

Kaj: Send fee reminders, attendance alerts, exam notices to parents via SMS (most reach 95%+ BD parents who don't check email).

Cost: ~৳0.25/SMS (Bangla Unicode ৳0.40). Minimum recharge: ৳1,000.

Steps:

1. Go to **https://www.sslwireless.com**

2. Contact sales at **sales@sslwireless.com** or call **+880 9612 334455**
3. Request a **Business SMS account** with:
   - Your company name (can be "Shikkha Platform" or your legal entity)
   - Trade license copy (or NID if sole proprietor)
   - Sender ID request (e.g., `SHIKKHA`, max 11 chars, alphanumeric)
4. Sign agreement + submit KYC docs
5. Pay initial recharge (minimum ৳1,000)
6. You'll get login credentials to `https://smsplus.sslwireless.com`
7. Inside dashboard → **Developer API** → Get:
   - **API Token** → `SSL_SMS_API_TOKEN`
   - **SID** → `SSL_SMS_SID`
   - **Approved Sender ID** → `SSL_SMS_SENDER_ID`
8. API endpoint: `https://smsplus.sslwireless.com/api/v3/send-sms` → `SSL_SMS_URL`

Timeline: ~3-5 working days for Sender ID approval from BTRC. **Alternative:** Alpha-SMS (https://sms.net.bd) — faster setup, similar pricing.

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5 ALPHA SMS (Fallback SMS Provider)

Kaj: Backup when SSL Wireless has downtime or for initial testing (no corporate KYC needed).

Cost: ৳0.25-0.40/SMS. Can start with ৳500 recharge.

Steps:

1. Go to **https://sms.net.bd** (Alpha SMS)
2. Sign up with phone number + NID
3. Recharge via bKash/Nagad (minimum ৳500)
4. Go to **API** section → Copy **API Key** → `ALPHA_SMS_API_KEY`
5. Request sender ID (standard templates available instantly, custom takes 2-3 days)
6. Copy approved Sender ID → `ALPHA_SMS_SENDER_ID`

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6 SSLCOMMERZ (Primary Payment Gateway)

Kaj: Accept payments via all Bangladeshi cards (VISA, MasterCard), bKash, Nagad, Rocket, Upay, DBBL Nexus, etc. — one integration, all methods.

Cost: 2.5% per successful transaction. No monthly fee.

Steps:

1. Go to https://sslcommerz.com
2. Click **"Merchant Registration"** → Fill form
3. Required docs:
   - Trade License (mandatory — sole proprietor works)
   - TIN Certificate
   - Bank account details (BD bank, in business name)
   - NID of proprietor/director
   - Website URL (must be live — not localhost)
   - Product/service description
4. Submit → Wait 3-7 working days for approval
5. Once approved, you'll get **Sandbox credentials** immediately for testing:
   - **Store ID** (sandbox) → `SSLCOMMERZ_STORE_ID`
   - **Store Password** (sandbox) → `SSLCOMMERZ_STORE_PASSWORD`
6. After testing + signing final agreement → you get **Live credentials**
7. Switch `SSLCOMMERZ_SANDBOX=false` before production

Gotcha: Your website must have Privacy Policy, Refund Policy, Terms & Conditions pages visible before they approve you. Have those ready.

7️⃣ BKASH DIRECT INTEGRATION (Optional, Lower Fees)

Kaj: Accept bKash payments directly without SSLCommerz middleman → lower fees (1.85% vs 2.5%).

Cost: 1.85% per transaction. No monthly fee. Requires business bKash merchant account.

Steps:

1. Contact **bKash Merchant Support**: **+880 9610011144** or visit https://developer.bka.sh
2. Apply for **bKash PGW (Payment Gateway)** — not personal/agent bKash
3. Required docs: Trade license, TIN, bank account (bKash-enabled), website URL
4. After approval (~7-14 days), you get:
   - **App Key** → `BKASH_APP_KEY`
   - **App Secret** → `BKASH_APP_SECRET`
   - **Username** → `BKASH_USERNAME`
   - **Password** → `BKASH_PASSWORD`
5. Sandbox URL: `https://tokenized.sandbox.bka.sh/v1.2.0-beta` → `BKASH_BASE_URL`
6. Production URL (after go-live): `https://tokenized.pay.bka.sh/v1.2.0-beta`

Tip: Start with SSLCommerz first (faster approval), add direct bKash later in Phase 3 when volume grows.

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8 NAGAD DIRECT INTEGRATION (Optional)

Kaj: Accept Nagad payments directly → lower fees than SSLCommerz passthrough.

Cost: 1.75% per transaction.

Steps:

1. Contact **Nagad Corporate**: merchant@nagad.com.bd or **16167**
2. Apply for **Nagad Payment Gateway**
3. Required: Same corporate KYC as above
4. After approval, you get:
   - **Merchant ID** → `NAGAD_MERCHANT_ID`
   - **Merchant Mobile Number** → `NAGAD_MERCHANT_NUMBER`
   - **Nagad Public Key** (for encryption) → `NAGAD_PG_PUBLIC_KEY`
   - **Merchant Private Key** (you generate this RSA keypair and share public with Nagad) → `NAGAD_MERCHANT_PRIVATE_KEY`
5. Base URL (sandbox): `https://sandbox.mynagad.com:10060/remote-payment-gateway-1.0`
6. Base URL (live): `https://api.mynagad.com/api/dfs` → `NAGAD_BASE_URL`

Gotcha: Nagad integration is more technical than bKash — requires RSA key pair generation. Use `openssl genrsa -out private.pem 2048` then share the `.pem` public portion with Nagad team.

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9 FIREBASE CLOUD MESSAGING (Push Notifications)

Kaj: Send push notifications to parent's phone/browser ("Your child was marked absent today") — free, real-time.

Cost: 100% free, unlimited.

Steps:

1. Go to **https://console.firebase.google.com**
2. Sign in with Google → **Add project**
3. Project name: `shikkha-platform-prod` → Disable Google Analytics (not needed for FCM) → Create

4. Once created, click **Web icon `</>`** to add a web app:
   - App nickname: `shikkha-web`
   - ✅ Enable Firebase Hosting (optional)
5. Firebase gives you a config object — copy each value:

   apiKey → NEXT_PUBLIC_FIREBASE_API_KEY

   authDomain → NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN

   projectId → NEXT_PUBLIC_FIREBASE_PROJECT_ID

   storageBucket → NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET

   messagingSenderId → NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID

   appId → NEXT_PUBLIC_FIREBASE_APP_ID

6. Go to **Project Settings** → **Cloud Messaging tab**:
   - Under **Web Push certificates**, click **Generate key pair**
   - Copy key → `NEXT_PUBLIC_FIREBASE_VAPID_KEY`
7. For server-side sending (Admin SDK):
   - Go to **Project Settings** → **Service accounts** → **Generate new private key**
   - Download JSON file
   - From JSON extract:
     - `project_id` → `FIREBASE_ADMIN_PROJECT_ID`
     - `client_email` → `FIREBASE_ADMIN_CLIENT_EMAIL`
     - `private_key` → `FIREBASE_ADMIN_PRIVATE_KEY` (keep the `\n` characters intact!)

Security: Never commit the downloaded JSON to git. Store private key in env var only.

🔟 CLOUDINARY (Image Optimization — Optional)

Kaj: Auto-resize student photos, transform on-the-fly (thumbnails for reports, watermarks for certificates), CDN delivery.

Cost: Free = 25GB storage + 25GB bandwidth + 25k transformations/month.

Steps:

1. Go to https://cloudinary.com → Sign up (free)
2. Dashboard shows your credentials immediately:

    - **Cloud name** → `CLOUDINARY_CLOUD_NAME`
    - **API Key** → `CLOUDINARY_API_KEY`
    - **API Secret** → `CLOUDINARY_API_SECRET`
3. Go to **Settings** → **Upload** → **Upload presets** → **Add upload preset**
4. Configure:
    - Name: `shikkha_unsigned` → `CLOUDINARY_UPLOAD_PRESET`
    - Signing Mode: `Unsigned` (for client-side uploads)
    - Folder: `shikkha/{school_id}/` (for multi-tenant organization)

Alternative: Skip Cloudinary entirely — Supabase Storage works fine for MVP. Only add Cloudinary if you need heavy image transformations.

1️⃣1️⃣ WHATSAPP BUSINESS CLOUD API (Optional but Powerful)

Kaj: Send fee reminders / attendance alerts via WhatsApp (higher open rate than SMS).

Cost: Per-conversation pricing: ৳0.40-0.80 per marketing msg, ৳0.20-0.40 for utility/transactional. Free tier: 1,000 conversations/month.

Steps:

1. Go to https://developers.facebook.com
2. Create a **Meta Business Account** (if you don't have one)
3. Create a new **App** → Type: **Business**
4. Add **WhatsApp product** → Get Started
5. Meta gives you a test number for free → verify with your phone
6. To send to real users, add your **Business Phone Number**:
    - Must be a new number, never used with personal WhatsApp
    - Display name must match your business
7. Verify business via Meta Business Verification (requires trade license, utility bill)
8. Once approved, collect:
    - **Phone Number ID** → `WHATSAPP_PHONE_NUMBER_ID`
    - **Business Account ID** → `WHATSAPP_BUSINESS_ACCOUNT_ID`
    - **Access Token** (permanent; generate via System User) → `WHATSAPP_ACCESS_TOKEN`
    - **App Secret** (for webhook verification) → `WHATSAPP_APP_SECRET`
    - You set `WHATSAPP_VERIFY_TOKEN` yourself (any random string for webhook validation)
9. Create **Message Templates** (required for outbound marketing):
    - Template: `fee_reminder_bn`, `attendance_absent_bn`, `exam_result_bn`
    - Submit for Meta approval (~24-48 hrs)

Phase: Add in Phase 4 after MVP proves. SMS first.

1️⃣2️⃣ ANTHROPIC CLAUDE API (AI Features)

Kaj: Generate smart report card comments, predict dropout risk, suggest fee structures, auto-translate between Bangla/English/Arabic.

Cost: Pay-per-use. Claude Sonnet 4.6: ~$3 per 1M input tokens, $15 per 1M output tokens. For ~500 students, ~$10-20/month.

Steps:

1. Go to **https://console.anthropic.com**
2. Sign up → Verify email + phone
3. Go to **Settings** → **API Keys** → **Create Key**
4. Name: `shikkha-platform-prod`
5. Copy (starts with `sk-ant-api03-`) → `ANTHROPIC_API_KEY`
6. Add payment method (credit card) at **Settings** → **Plans & Billing**
7. Set monthly spending limit (e.g., $50/mo) to prevent runaway costs
8. Model: `claude-sonnet-4-6` → `ANTHROPIC_MODEL`

Tip: Start with free API credits (usually $5 on signup). Track usage at **Usage** tab daily for first week.

1️⃣3️⃣ OPENAI (Backup AI — Optional)

Kaj: Fallback if Anthropic is down, or for features better served by OpenAI (e.g., Whisper for audio transcription of parent-teacher conferences).

Cost: GPT-4o-mini: $0.15/1M input, $0.60/1M output (very cheap).

Steps:

1. Go to **https://platform.openai.com**
2. Sign up → Verify
3. Go to **API Keys** → **Create new secret key**
4. Copy (starts with `sk-proj-`) → `OPENAI_API_KEY`
5. Add payment method + set monthly limit

1️⃣4️⃣ SENTRY (Error Tracking)

Kaj: Catch bugs in production before users complain. See stack traces, user context, replay sessions.

Cost: Free = 5,000 errors/month + 50 replays. Team plan $26/mo for growth.

Steps:

1. Go to **https://sentry.io** → Sign up
2. Create organization: `shikkha` → `SENTRY_ORG`
3. Create project:
   - Platform: **Next.js**
   - Name: `shikkha-platform` → `SENTRY_PROJECT`
4. Sentry shows you the DSN → copy → `NEXT_PUBLIC_SENTRY_DSN`
5. Go to **Settings** → **Auth Tokens** → **Create New Token**
6. Scopes needed: `project:read`, `project:releases`, `org:read`
7. Copy token → `SENTRY_AUTH_TOKEN` (used during build for sourcemap upload)

1️⃣5️⃣ POSTHOG (Product Analytics)

Kaj: Track which features users actually use, funnel analysis (signup → first mark entry → export report), session replay.

Cost: Free = 1M events/month + session recording. $0.00031/event after.

Steps:

1. Go to **https://posthog.com** → Sign up (EU Cloud or US Cloud)
2. Create project: `shikkha-platform`
3. Go to **Project Settings** → **Project API Key**
4. Copy → `NEXT_PUBLIC_POSTHOG_KEY`
5. Host URL:
   - US Cloud: `https://us.i.posthog.com`
   - EU Cloud: `https://eu.i.posthog.com`
   - Set as `NEXT_PUBLIC_POSTHOG_HOST`

1️⃣6️⃣ UPSTASH REDIS (Caching + Rate Limiting)

Kaj: Cache frequently-read data (school config, permission lists) + rate limit login attempts / API calls to prevent abuse.

Cost: Free = 10,000 requests/day + 256MB storage. Pay-as-you-go $0.2/100k requests after.

Steps:

1. Go to **https://upstash.com** → Sign up with GitHub
2. Click **Create Database**
3. Name: `shikkha-cache`, Type: **Regional**, Region: **AP-Southeast-1 (Singapore)**
4. Enable **TLS**
5. Once created, go to database details page:
    - **REST API URL** → `UPSTASH_REDIS_REST_URL`
    - **REST API Token** → `UPSTASH_REDIS_REST_TOKEN`

1️⃣7️⃣ GOOGLE MAPS API (Location Features)

Kaj: School location picker during onboarding, transport route visualization, parent home-address geocoding.

Cost: $200 free credit/month (≈ 28,500 map loads free).

Steps:

1. Go to **https://console.cloud.google.com**
2. Sign in → Create project: `shikkha-platform`
3. Go to **APIs & Services** → **Library** → Enable these APIs:
    - Maps JavaScript API
    - Geocoding API
    - Places API
    - Directions API (for transport routes)
4. Go to **Credentials** → **Create credentials** → **API Key**
5. Copy key → `NEXT_PUBLIC_GOOGLE_MAPS_API_KEY`
6. **IMPORTANT:** Click on the key → **Application restrictions** → **HTTP referrers** → Add:
    - `http://localhost:3000/*`
    - `https://shikkha.app/*` (or your domain)
    - `https://*.vercel.app/*`
7. Set **API restrictions** → Select only the 4 APIs you enabled (prevents abuse)
8. Set a **billing cap** at $5/mo to prevent accidental overrun

---

1⃣8⃣ CRON / QSTASH (Scheduled Jobs)

Kaj: Run scheduled jobs — daily attendance SMS summary at 9 PM, monthly fee generation on 1st of month, overdue fee reminder on 25th.

Option A: Vercel Cron (free, 2 crons on Hobby plan) — just use `CRON_SECRET` (you generate).
Option B: Upstash QStash (better for multi-step jobs, $0.5/100k invocations).

For Vercel Cron:

1. Generate secret: `openssl rand -hex 32`
2. Set as `CRON_SECRET`
3. In `vercel.json`, configure:

```
{
  "crons": [
    {"path": "/api/cron/daily-attendance-sms", "schedule": "0 15 * * *"},
    {"path": "/api/cron/monthly-fee-gen", "schedule": "0 0 1 * *"}
  ]
}
```

For QStash:

1. Go to **https://console.upstash.com/qstash**
2. Copy:
   - **Token** → `QSTASH_TOKEN`
   - **Current Signing Key** → `QSTASH_CURRENT_SIGNING_KEY`
   - **Next Signing Key** → `QSTASH_NEXT_SIGNING_KEY` (for key rotation)

1⃣9⃣ DOMAIN (You Need This for Email + Branding)

Kaj: Your brand URL (e.g., `shikkha.app`, `bidyalaya.com.bd`).

Cost:

- `.com` / `.app`: ~$12-15/year (Namecheap, Porkbar)

- `.com.bd` / `.edu.bd`: ~৳3,000/year (via BTCL or authorized registrars)

Steps:

1. Search availability at **https://www.namecheap.com** or **https://porkbun.com**
2. Buy domain → 2-factor auth on registrar account
3. Point DNS to Vercel:
    - Add A record: `@` → `76.76.21.21`
    - Add CNAME: `www` → `cname.vercel-dns.com`
4. Verify in Vercel Dashboard → Settings → Domains
5. Add Resend DNS records (TXT for SPF/DKIM/DMARC) — covered in Resend step above

.com.bd Domain: Need trade license + NID. Register via **https://btcl.com.bd** or authorized resellers like Exonhost, HostBangla.

2⃣0⃣ JWT & SESSION SECRETS (Generate Locally — No Service)

Steps:

Run these in terminal and paste output directly:

```
# JWT_SECRET

openssl rand -base64 32

# SESSION_SECRET

openssl rand -hex 32

# CRON_SECRET

openssl rand -hex 32
```

Rule: Use **different values** for dev vs production. Regenerate if ever leaked.

📦 Summary — Required vs Optional Checklist

✅ Absolute Must-Have (MVP launch, Phase 0-2)

- ☐ Supabase (URL, anon key, service role key, JWT secret, DB password)
- ☐ Vercel (account + domain connected)
- ☐ Resend (API key + verified domain)
- ☐ SSL Wireless or Alpha SMS (API key + sender ID)
- ☐ SSLCommerz (store ID + password)
- ☐ Domain (bought + DNS configured)
- ☐ JWT_SECRET + SESSION_SECRET (locally generated)
- ☐ SUPER_ADMIN_EMAIL + initial password (for first login)

⚠️ Recommended (Phase 3-4)

- ☐ Firebase FCM (push notifications — free)
- ☐ Sentry (error tracking — free)
- ☐ PostHog (analytics — free)
- ☐ Upstash Redis (caching — free)
- ☐ bKash direct (lower fees)

🔵 Nice-to-Have (Phase 5+)

- ☐ Anthropic Claude API (AI features)
- ☐ Cloudinary (only if heavy image ops)
- ☐ WhatsApp Business API
- ☐ Nagad direct
- ☐ Google Maps API
- ☐ QStash (advanced scheduling)

❌ Skip Unless Needed

- Stripe (only for international)
- OpenAI (only if Anthropic fails or need Whisper)

🛡️ Security Rules — Protect Your Keys

1. **Never commit `.env.local` to git.** Add to `.gitignore` (Next.js does this by default).

2. **Use different keys for dev vs production.** Two Supabase projects, two Resend domains, etc.
3. **Rotate secrets every 90 days.** Set a reminder.
4. `NEXT_PUBLIC_*` **keys are exposed to browser** — never put sensitive keys with this prefix.
5. **Service role keys / API secrets / DB passwords** → ONLY used in `app/api/` routes, server actions, edge functions. Never in client components.
6. **Use Vercel's encrypted env vars** for production — don't paste into their UI from shared screen.
7. **If a key leaks:** rotate immediately at the source, redeploy, audit access logs.
8. **Set spending caps** on every pay-per-use service (Anthropic, Google Maps, OpenAI).

📅 Timeline — Realistic Setup Schedule

Day	Task	Time
Day 1	Supabase + Vercel + Domain + Resend	2 hrs
Day 2	Alpha SMS quick signup, test SMS send	1 hr
Day 3-7	SSL Wireless application + Sender ID approval (wait for BTRC)	Passive
Day 5-10	SSLCommerz application (wait for approval)	Passive
Week 2	Firebase FCM + Sentry + PostHog	2 hrs
Week 3	Upstash + Google Maps + generate secrets	1 hr
Month 2	bKash direct + WhatsApp Business	As needed
Month 3	Anthropic AI + Cloudinary	As needed

Total hands-on setup time: ~8 hours. Most time is waiting for BD service approvals.

🆘 Hand This to Claude Code — What to Say

> *"Reference `ENV_SETUP_GUIDE.md` for the complete list of environment variables. Use these exact variable names in all code. For each integration (Supabase, SSLCommerz, SMS, etc.), create a typed config loader in `lib/config/` that validates required env vars at app startup using `zod`. Fail fast if any required var is missing in production. For feature flags, read `NEXT_PUBLIC_ENABLE_*` and gate UI + API routes accordingly so we can ship MVP without all integrations ready."*

Pro-tip: Create a `env.schema.ts` file early with zod validation — Claude Code will reference this to know which env vars are required vs optional:

```
// lib/config/env.schema.ts

import { z } from 'zod';

export const envSchema = z.object({

  // Required

  NEXT_PUBLIC_SUPABASE_URL: z.string().url(),

  NEXT_PUBLIC_SUPABASE_ANON_KEY: z.string().min(1),

  SUPABASE_SERVICE_ROLE_KEY: z.string().min(1),

  RESEND_API_KEY: z.string().startsWith('re_'),

  // Optional

  ANTHROPIC_API_KEY: z.string().optional(),

  WHATSAPP_ACCESS_TOKEN: z.string().optional(),

  // ...

});
```

This file alone will save hours of debugging.